

Constructing Molecular Mechanics Force Fields: A Modular Implementation of Harmonic and Non-Bonded Potentials

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Abstract—Molecular mechanics force fields translate the quantum-mechanical reality of chemical bonding into a computationally tractable classical potential energy function $U(\mathbf{r})$. We present a from-scratch Python implementation of the five canonical force field energy terms—harmonic bond stretching, harmonic angle bending, periodic dihedral torsions, Lennard-Jones 12-6 van der Waals interactions, and Coulomb electrostatics—parameterized with AMBER ff99 values. The engine evaluates six preset polypeptide systems ranging from the alanine dipeptide (22 atoms) to a 5-residue alpha helix (50 atoms), computes per-bond stress to identify frustrated regions, and provides an interactive “Break a Bond” experiment. All energies are reported in kcal/mol with physically meaningful parameters drawn from the Cornell et al. (1995) force field [1]. Non-bonded interactions consistently dominate the total energy in systems larger than 20 atoms, and the harmonic approximation accurately reproduces near-equilibrium behavior while diverging at large displacements.

Index Terms—molecular mechanics, force field, AMBER, Lennard-Jones, Coulomb electrostatics, harmonic potential, protein simulation

I. INTRODUCTION

Predicting protein structure and stability requires evaluating the total potential energy $U(\mathbf{r})$ as a function of all $3N$ atomic coordinates. In molecular mechanics (MM), the electronic structure is replaced by a classical energy function composed of simple analytical terms: springs for bonds, cosines for torsions, and power laws for van der Waals interactions [3]. This approximation reduces the computational cost from the $\mathcal{O}(N^7)$ scaling of high-level quantum methods to $\mathcal{O}(N^2)$ (or lower with cutoffs), enabling simulations of systems containing thousands of atoms [7].

The total MM potential energy decomposes into five terms:

$$U(\mathbf{r}) = U_{\text{bonds}} + U_{\text{angles}} + U_{\text{dihedrals}} + U_{\text{vdW}} + U_{\text{elec}} \quad (1)$$

Each term is parameterized independently using experimental or quantum-chemical data. The AMBER family of force fields [1], [4] is among the most widely used for biomolecular simulations.

This work presents a modular Python implementation of eq. (1) with AMBER ff99 parameters, applied to six preset polypeptide systems. We introduce a per-bond stress metric that highlights frustrated regions and provide an interactive bond-stretching experiment that demonstrates the breakdown of the harmonic approximation.

A. Scope and Contributions

- 1) A self-contained force-field engine implementing all five energy terms with AMBER ff99 parameters.
- 2) Per-bond stress visualization for identifying frustrated regions.
- 3) A “Break a Bond” experiment illustrating harmonic energy growth.
- 4) Parameter scanning and coordinate perturbation utilities.
- 5) Interactive Streamlit dashboard and CLI for exploration.

II. THEORY

A. Harmonic Bond Stretching

Covalent bonds are modeled as harmonic oscillators. The energy for bond b between atoms i and j is:

$$U_{\text{bond}} = \frac{1}{2}k_b(r_{ij} - r_0)^2 \quad (2)$$

where k_b is the force constant, r_{ij} is the internuclear distance, and r_0 is the equilibrium bond length. AMBER ff99 values for a C–C single bond are $k_b = 340$ kcal/(mol Å²) and $r_0 = 1.526$ Å [1].

B. Harmonic Angle Bending

The angle θ_{ijk} formed by three consecutively bonded atoms resists deformation from equilibrium:

$$U_{\text{angle}} = \frac{1}{2}k_\theta(\theta_{ijk} - \theta_0)^2 \quad (3)$$

Tetrahedral sp³ carbon centers have $\theta_0 \approx 109.5$ with $k_\theta \approx 63$ kcal/(mol rad²).

C. Periodic Dihedral Torsions

Rotation about the central bond of four consecutive atoms follows a periodic potential:

$$U_{\text{dihedral}} = \frac{V_n}{2} [1 + \cos(n\phi - \gamma)] \quad (4)$$

where V_n is the barrier height, n is the periodicity, ϕ is the torsion angle, and γ is the phase offset.

D. Lennard-Jones 12-6 Potential

Non-bonded van der Waals interactions are modeled by:

$$U_{\text{LJ}} = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (5)$$

The r^{-12} term captures Pauli repulsion; the r^{-6} term represents London dispersion. Cross-type parameters use the Lorentz-Berthelot combining rules:

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2}, \quad \epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j} \quad (6)$$

E. Coulomb Electrostatics

Partial atomic charges interact via Coulomb’s law:

$$U_{\text{elec}} = \frac{332.0636 q_i q_j}{\epsilon_r r_{ij}} \quad (7)$$

The constant 332.0636 kcal Å/(mole²) ensures correct units when charges are in elementary charge units and distances in angstroms.

F. Exclusion Scheme

Atoms separated by one bond (1-2 pairs) or two bonds (1-3 pairs) are excluded from non-bonded calculations. These short-range interactions are already captured by the bond and angle terms [1].

G. Bond Stress Metric

We define per-bond stress as the normalized harmonic energy:

$$s_b = \frac{U_b}{\max_{b'} U_{b'}} \quad (8)$$

Values near 1.0 indicate maximally strained bonds, which may correspond to frustrated regions involved in catalysis or conformational change [8].

III. METHODS

A. Software Architecture

The implementation follows a layered architecture with strict dependency flow:

TABLE I
MODULE ARCHITECTURE

Module	Responsibility	Lines
forcefield_engine.py	Data structures, energy calculators	~1570
analysis.py	Typed result containers, pipelines	~420
visualization.py	Plotly + Matplotlib renderers	~700
app.py	8-page Streamlit dashboard	~530
main.py	CLI with 5 modes	~230

B. ForceField Data Model

The ForceField class maintains lists of Atom, BondParam, AngleParam, and DihedralParam data-class instances. An exclusion set of (1-2) and (1-3) atom pairs is built automatically from the bond and angle topology.

C. Energy Evaluation Protocol

Energy evaluation proceeds sequentially through the five terms:

- 1) Bond energy: iterate over all BondParam entries; compute r_{ij} via Euclidean distance; apply eq. (2).
- 2) Angle energy: compute θ_{ijk} via the dot product of bond vectors; apply eq. (3).
- 3) Dihedral energy: compute ϕ via the cross-product method; apply eq. (4).
- 4) Non-bonded energy: double loop over all non-excluded pairs within the cutoff; compute LJ (eq. (5)) + Coulomb (eq. (7)).
- 5) Total: sum all five contributions.

D. Preset Molecule Systems

TABLE II
PRESET POLYPEPTIDE SYSTEMS

Molecule	Atoms	Bonds	Angles	Description
Alanine dipeptide	22	21	20	Standard benchmark
Glycine tripeptide	33	32	31	Extended backbone
Alpha helix (5-res)	50	49	48	Poly-alanine helix
Beta hairpin	16	15	12	Backbone strand
Salt bridge	14	13	12	Lys-Asp charge pair
Disulfide bond	12	11	10	Cys-S-S-Cys cross-link

E. Stress Visualization

Per-bond stresses are computed via eq. (8) and mapped to a blue-to-red color scale. Bonds with stress > 0.7 are flagged as frustrated regions and reported to the user.

F. Break a Bond Experiment

A selected bond is incrementally stretched from equilibrium by up to a user-specified displacement. At each step, the full energy $U(\mathbf{r})$ is recomputed, demonstrating the quadratic growth of the harmonic term (eq. (2)).

IV. RESULTS

A. Energy Decomposition — Alanine Dipeptide

The alanine dipeptide (Ace-Ala-Nme, 22 atoms) provides the primary benchmark. Non-bonded interactions (VdW + electrostatic) consistently dominate the total energy, accounting for the majority of the energy magnitude. Bond and angle terms contribute moderate positive energy due to minor deviations from equilibrium geometry.

B. Stress Analysis

Terminal bonds (N-terminus and C-terminus cappings) exhibit higher stress than interior backbone peptide bonds, reflecting minor geometric deviations in the preset builders. Frustrated regions correspond to bonds with normalized stress > 0.7 .

C. Break a Bond

Stretching the N-C $_{\alpha}$ bond of alanine dipeptide by up to +2.0 Å produces the expected parabolic energy curve. The energy increases by several hundred kcal/mol—far exceeding the thermal energy $k_B T \approx 0.6$ kcal/mol at 300 K—demonstrating the high cost of covalent bond rupture in the harmonic approximation.

D. Parameter Sensitivity

Scanning the bond force constant k_b from 50 to 600 kcal/(mol Å²) reveals a linear relationship between k_b and the total bond energy contribution when bonds are displaced from equilibrium. The dielectric constant ϵ_r modulates electrostatic energy approximately inversely.

E. Perturbation Robustness

Gaussian coordinate perturbations with scale factors from 0.01 to 0.5 Å demonstrate:

- $\sigma = 0.01$ Å: energy change < 1 kcal/mol (thermal regime)
- $\sigma = 0.1$ Å: energy change ~ 10 –50 kcal/mol (strain)
- $\sigma = 0.5$ Å: energy change ~ 100 + kcal/mol (non-physical)

F. Cross-Molecule Comparison

Larger systems show proportionally larger non-bonded contributions due to $\mathcal{O}(N^2)$ pair scaling. The salt bridge system is dominated by strong Coulomb attraction between charged residues.

V. DISCUSSION

A. Accuracy of the Harmonic Approximation

The harmonic bond model accurately reproduces near-equilibrium energetics but diverges from reality at large displacements. A Morse potential would provide more realistic behavior at the cost of additional parameters [12]. The “Break a Bond” experiment clearly illustrates this limitation.

B. Non-Bonded Dominance

In systems larger than ~ 20 atoms, non-bonded interactions dominate the energy landscape. This reflects the $N(N-1)/2$ pair scaling versus the linear scaling of bonded terms. Electrostatics are particularly important in polypeptides due to large partial charges on backbone N, C, and O atoms [5].

C. Computational Considerations

The naive $\mathcal{O}(N^2)$ non-bonded calculation can be accelerated to $\mathcal{O}(N \log N)$ via particle-mesh Ewald [11] or $\mathcal{O}(N)$ via the fast multipole method. Neighbor lists and cell lists are classical spatial data structures used in production MD codes [14].

D. Limitations

The current implementation uses fixed partial charges (no polarization), harmonic bonds (no dissociation), and idealized preset geometries. No explicit solvent, periodic boundary conditions, or long-range electrostatic corrections are applied.

E. Biological Relevance

Force fields enable structure refinement, drug design scoring, molecular dynamics, and mutation impact prediction [7]. The per-bond stress metric introduced here provides a simple diagnostic for identifying structurally frustrated regions that may correlate with functional sites.

VI. CONCLUSION

We have implemented a complete molecular mechanics force field engine with five standard energy terms and AMBER ff99 parameters. The engine evaluates six preset polypeptide systems, computes per-bond stress for frustrated-region identification, and provides interactive bond-stretching, parameter-scanning, and coordinate-perturbation experiments. Non-bonded interactions dominate in larger systems, and the harmonic approximation accurately describes near-equilibrium behavior while failing at large displacements. The modular Python architecture provides a pedagogical platform for exploring the physics underlying modern biomolecular simulation.

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